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Asymmetric bifurcations in a pressurised circular thin plate under initial tension

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ABSTRACT

It has been known for some time that under certain circumstances the axisymmetric solution describing the deformation experienced by a stretched circular thin plate or membrane under sufficiently strong normal pressure does not represent an energy-minimum configuration. By using the method of adjacent equilibrium a set of coordinate-free bifurcation equations is derived here by adopting the Föppl–von Kármán plate theory. A particular class of asymmetric bifurcation solutions is then investigated by reduction to a system of ordinary differential equations with variable coefficients. The localised character of the eigenmodes is confirmed numerically and we also look briefly at the role played by the background tension on this phenomenon.

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1. Introduction

Friedrichs and Stoker (1941) were the first to show, in what later turned out to be a landmark paper in the field of elastic stability, the coexistence of tensile and compressive stresses in thin flat plates subjected to uniform compressive in-plane loads. By considering a circular geometry in conjunction with the Föppl–von Kármán (FvK) equations, their boundary-layer calculations predicted that the compressive stresses applied at the edge of the plate may result in tension at the centre due to buckling, and thus suggested that the plate experienced stretching in that vicinity. Since their analysis was based on radially symmetric buckling, the existence of compressive stresses near the outer rim opened the possibility of secondary bifurcations characterised by circumferential wrinkling confined near that edge; this expectation was only later confirmed in the numerical work of Cheo and Reiss (1973).

The related situation of nonlinear bending for circular thin elastic plates under transverse pressure was discussed by Bromberg (1956) by using the same boundary-layer techniques as in the earlier analyses of Friedrichs and Stoker. As pointed out by Yanowitch (1956) around the same time, for certain types of boundary conditions, Bromberg's analysis ceases to be valid because of the imminent danger of circumferential buckling associated with the existence of negative azimuthal stresses near the plate boundary. All these analyses were based on the FvK equations, but it is well known that this behaviour persists for Reissner's large deflection plate model as well (e.g., see Jahsman et al., 1962).

Panov and Feodosiev (1948) seem to be the first to consider in some detail the possibility of buckling of pressurised thin elastic plates (cf. Timoshenko and Gere, 1961, footnote on page 402). Their analysis was conducted using a special theory of shallow shells and employing a very simple Galerkin procedure. As pointed out by Cheo and Reiss (1973) the oversimplified choice of ansatz led those authors to results which differ considerably from other relevant data available in the literature. Much later a number of extensive numerical contributions were made by Shilkrut (1983, 2002) and Shilkrut and Kovachi (1986), whose interest was primarily in pressurised spherical elastic caps; the elastoplastic counterpart of this situation was discussed by Triantafyllidis (1984) by using more rigorous analytical tools. In more recent time the problem has also been revisited by Adams (1993). He investigated an elastic plate uniformly stretched radially prior to the transverse load being applied. In his analysis a nonlinear axisymmetric pre-bifurcation state is adopted, and the FvK equations are then solved for a concentrated load applied at the centre of the plate. The non-dimensional equations involve only the Poisson's ratio of the elastic material, the rescaled version of the background tension N_0 and the applied pressure P. Rather curiously, it was discovered that the buckling patterns tend to be confined near the outer half of the plate, even for low or zero values of N_0; increasing this parameter resulted in the eigenmodes shifting even further towards the outer rim, but this trend was very mild (see Fig. 7 in Adams, 1993) and the mechanism by which the eigenmodes became localised was left unexplored. It is the purpose of the present work to look more carefully at a modification of that problem in which the concentrated load is replaced by a uniform pressure applied over the upper face of the plate.

Our interest in this problem has been motivated by the recent investigations of Coman (2008) and Coman and Bassom (2007a,b, 2008), who explored the *linear* bifurcations for several pre-buckling deformations displaying regions of both tensile and compressive stresses. The situation was different from that covered by Friedrichs

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and Stoker, in the sense that these mixed stresses were present before the actual bifurcation took place and nonlinearities were not accounted for. Furthermore, all problems discussed therein depended in a crucial way on one form or another of initial tension that was shown to be responsible for the presence of a large dimensionless parameter in the rescaled buckling equations. This important feature was exploited in developing a general boundary-layer strategy that turned out to be capable of capturing both the dependence of the critical load and the number of wrinkles on the asymptotic parameter. Except for the isolated scenario of twisted thin elastic strips reported by Coman and Bassom (2008), these studies were conducted under the assumption that the plates were loaded in their plane, so the pre-buckling state was one of plane stress. The situation described by Cheo and Reiss (1973) or Adams (1993) is quite different, in that the initial deformation of the plate involves a non-zero transverse displacement (hence the pre-buckling rotations can not be ignored). Also, since the basic state must be described by a nonlinear boundary-value problem, the dependence of the buckling equations on the eigenvalue is *nonlinear*. A yet another consequence brought forth by the three-dimensional character of the basic state is that now the usual fourth-order buckling equation for the transverse deflection will have to be supplemented by two extra equations for the in-plane displacement fields. The result is a fairly complicated eight-order system of partial differential equations with variable coefficients which, incidentally, are not available in closed form.

The plan of the paper is as follows. In the next section we recapitulate briefly the equations for the radially symmetric pre-buckling state and identify the non-dimensional combinations of parameters that control the behaviour of the pressurised thin plate. A new coordinate-free formulation of the FvK bifurcation equations is reported in Section 3. For circular geometry a normal mode solution reduces them to three coupled ordinary differential equations involving the arbitrary mode number proportional to the number of regular wrinkles around the circumference of the plate. In general, this quantity must be determined numerically by minimising the load with respect to it, and constitutes one of the main challenges in problems of asymmetric buckling. To appreciate the effect of localisation and assess the possible use of asymptotic techniques that will be reported elsewhere (Coman, in preparation), a number of numerical simulations are carried out in Section 4. We focus on the role played by the background tension and give some heuristic explanations for the behaviour observed in numerical simulations.

2. Basic state

Let $a>0$ be the radius of a thin elastic plate of thickness h ($a/h \ll 1$), which is radially stretched by applying the uniform in-plane tension $N_0 \equiv \sigma h$ along its circumference. Here $\sigma>0$ is the applied constant stress acting in the mid-plane of the plate. The linearly elastic material is characterised by the Young's modulus E and Poisson's ratio ν. It is assumed that this configuration is further subjected to a uniform pressure $p>0$ normal to the plane of the plate – see Fig. 1. Under the assumption of axial symmetry the problem has been solved by a number of investigators (a comprehensive review is given by Chia, 1980). For the sake of completeness, we include here a brief overview of those aspects that are relevant to the later parts of the paper.

Let $w=\mathring{w}(r)$ and $u=\mathring{u}(r)$ be the transverse and, respectively, the in-plane displacements of the plate, with $0 \le r \le a$ denoting the distance from its centre. The Föppl–von Kármán (FvK) nonlinear equations applicable to this situation are

$$D\nabla^4 w - \nabla\cdot(\boldsymbol{N}\cdot\nabla w) = p, \tag{1a}$$

$$\nabla\cdot\boldsymbol{N} = \boldsymbol{0}, \tag{1b}$$

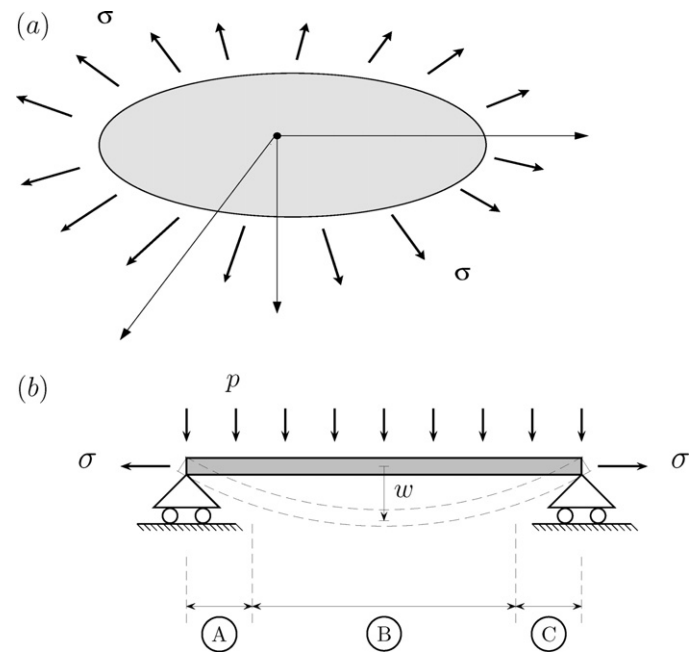

Fig. 1. The stretched thin film shown in (a) is subjected to normal pressure (treated here as dead loading) – see (b). For a critical combination of p and σ, compressive stresses develop in a ring-like region near the outer rim; $\mathring{N}_{\theta\theta}(r)<0$ in Ⓐ and Ⓒ, while in the region Ⓑ the azimuthal stresses are tensile.

where $D \equiv Eh^3/12(1-\nu^2)$ represents the bending rigidity of the plate and $\boldsymbol{N}=(\mathring{N}^{\alpha\beta})$ ($\alpha,\ \beta \in \{1,\ 2\}$) is the membrane stress tensor; the axial symmetry assumption requires that $\mathring{N}^{\alpha\beta} \equiv \mathring{N}^{\alpha\beta}(r)$. In general, (1) can be reduced to a system of two second-order equations for the slope $d\mathring{w}/dr$ and the in-plane displacement $\mathring{u}$ (e.g., see Chia, 1980). However, since the bifurcation equations derived in the next section will contain the slope as well as the radial and azimuthal stresses of the basic state, it seems more appropriate to derive directly equations for these quantities, which is what we do next.

As already mentioned the situation we have in mind involves a stretched thin plate subjected also to a normal pressure. It helps to think of this scenario as consisting of two separate phases. Initially the plate is stretched by the in-plane uniform load N_0. The solution corresponding to this first stage is found by taking $w \equiv 0$ and $P=0$ in (1); hence

$$N_{rr}(r) = N_{\theta\theta}(r) \equiv N_0 \qquad \text{and} \qquad u(r) = \frac{N_0}{Eh}(1-\nu)r.$$

After the pressure is applied, the in-plane state of stress is expressed as $\mathring{N}_{rr}(r) = N_0 + \widehat{N}_{rr}(r)$ and $\mathring{N}_{\theta\theta}(r) = N_0 + \widehat{N}_{\theta\theta}(r)$, where $\widehat{N}_{jj}$ ($j \in \{r,\ \theta\}$) represent the incremental changes in the corresponding components of the membrane tensor due to the applied pressure. It is found useful to adopt the following non-dimensional variables/parameters:

$$\rho := \frac{r}{a}, \qquad \mathring{w}^* = \frac{\mathring{w}}{h}, \quad \widehat{N}^*_{rr} := (1-\nu^2)\frac{\widehat{N}_{rr}}{Eh}\left(\frac{a}{h}\right)^2,$$

$$\widehat{N}^*_{\theta\theta} := (1-\nu^2)\frac{\widehat{N}_{\theta\theta}}{Eh}\left(\frac{a}{h}\right)^2, \tag{2a}$$

$$\mu^2 := (1-\nu^2)\left(\frac{\sigma}{E}\right)\left(\frac{a}{h}\right)^2, \quad P := (1-\nu^2)\left(\frac{p}{E}\right)\left(\frac{a}{h}\right)^4,$$

$$\Theta := \frac{d\mathring{w}^*}{d\rho}. \tag{2b}$$

After dropping the asterisks for notational convenience, routine manipulations (see Sheplak and Dugundji, 1998 for a traditional derivation without tensors) lead to the nonlinear differential equations satisfied by the unknown non-dimensional functions $\widehat{N}_{rr}(\rho)$, $\widehat{N}_{\theta\theta}(\rho)$, $\Theta(\rho)$ in $0<\rho<1$,

$$\frac{d\widehat{N}_{rr}}{d\rho}+\frac{1}{\rho}(\widehat{N}_{rr}-\widehat{N}_{\theta\theta})=0, \tag{3a}$$

$$\frac{d\widehat{N}_{\theta\theta}}{d\rho}-\frac{1}{\rho}(\widehat{N}_{rr}-\widehat{N}_{\theta\theta})=-(1-\nu^2)\frac{\Theta^2}{2\rho}, \tag{3b}$$

$$\frac{d^2\Theta}{d\rho^2}+\frac{1}{\rho}\frac{d\Theta}{d\rho}-\left(12\mu^2+\frac{1}{\rho^2}\right)\Theta-12\widehat{N}_{rr}\Theta=6\rho P. \tag{3c}$$

A number of different edge restraints can be considered, although it must be kept in mind that buckling requires that $\widehat{N}_{\theta\theta}(\rho)<-\mu^2$ for at least one value of $\rho\in(0,1]$. Previous experience – see Cheo and Reiss (1973), Adams (1993) and Chia (1980) – has shown that partial clamp and roller-type constraints satisfy this requirement. In the interest of brevity we shall consider only the former set of boundary conditions. Thus, the equations in (3) will be solved subject to the requirements that

$$\Theta(1)=0,\qquad \widehat{N}_{rr}(1)=0, \tag{4}$$

together with the regularity conditions at the centre of the plate,

$$\Theta(0)=0,\qquad \widehat{N}_{rr}(0)=\widehat{N}_{\theta\theta}(0). \tag{5}$$

We mention in passing that the pre-buckling displacement fields are given by the formulae

$$\mathring{w}^*(\rho)=\int_0^\rho \Theta(s)\,ds,\qquad \mathring{u}^*(\rho)=\frac{\rho[\widehat{N}_{\theta\theta}(\rho)-\nu\widehat{N}_{rr}(\rho)]}{1-\nu^2},$$

where $\mathring{u}^*:=\mathring{u}a/h^2$ represents the non-dimensional radial displacement field.

The diagonal membrane tensor whose components are the functions $\widehat{N}_{rr}$ and $\widehat{N}_{\theta\theta}$ obtained by solving (3) and (4) will be referred to as $\widehat{\boldsymbol{N}}$. Note that the total state of stress in the stretched and pressurised thin plate will be $\mu^2\boldsymbol{I}+\widehat{\boldsymbol{N}}$, where $\boldsymbol{I}$ is the two-dimensional identity tensor.

3. The bifurcation equations

The novel derivation presented below is fairly general and includes as particular cases both rectangular and circular plates. To fix ideas we shall assume that we have available a pair of reciprocal bases $\{\boldsymbol{g}_i\}$ and $\{\boldsymbol{g}^i\}$ ($i=1,2,3$), respectively, adapted for the problem at hand; $\boldsymbol{g}_3$ is a unit vector perpendicular to the plate mid-plane. Expressed in the latter base, the contravariant components of the two-dimensional identity tensor $\boldsymbol{I}$ are $g^{\alpha\beta}\equiv\boldsymbol{g}^\alpha\cdot\boldsymbol{g}^\beta$ ($\alpha,\beta=1,2$).

We shall make use again of the FvK equations (1), and start by observing that the first equation (1a) can be simplified. Indeed, recalling (1b) and using a simple tensor calculus identity yields $\nabla\cdot(\boldsymbol{N}\cdot\nabla w)=(\nabla\cdot\boldsymbol{N})\cdot\nabla w+\boldsymbol{N}:(\nabla\otimes\nabla w)=\boldsymbol{N}:(\nabla\otimes\nabla w)$. The membrane tensor $\boldsymbol{N}=(N^{\alpha\beta})$ that enters in (1) has the expression $N^{\alpha\beta}=C[(1-\nu)n^{\alpha\beta}+\nu\, n\, g^{\alpha\beta}]$, where $C\equiv Eh/(1-\nu^2)$ represents the in-plane stiffness and

$$\boldsymbol{n}=(n^{\alpha\beta})=\frac{1}{2}(\boldsymbol{u}\otimes\nabla+\nabla\otimes\boldsymbol{u})+\frac{1}{2}(\nabla w)\otimes(\nabla w), \tag{6a}$$

$$n\equiv n^\gamma_\gamma=\boldsymbol{n}:\boldsymbol{I}=\nabla\cdot\boldsymbol{u}+\frac{1}{2}|\nabla w|^2, \tag{6b}$$

here $\boldsymbol{u}\equiv u_1\,\boldsymbol{g}^1+u_2\,\boldsymbol{g}^2$ denotes the in-plane displacement of the plate mid-plane and w is the transverse displacement along $\boldsymbol{g}^3$. With the help of (6) we find that

$$\boldsymbol{N}=C\left\{\left(\frac{1-\nu}{2}\right)[\nabla\otimes\boldsymbol{u}+\boldsymbol{u}\otimes\nabla+(\nabla w)\otimes(\nabla w)]+\nu\left(\nabla\cdot\boldsymbol{u}+\frac{1}{2}|\nabla w|^2\right)\boldsymbol{I}\right\}. \tag{7}$$

The relevant bifurcation equations are obtained by employing the method of adjacent equilibrium. For short, in our particular context this amounts to writing the equilibrium equations on two neighbouring stressed configurations and taking their difference, followed by a geometrical linearisation of the kinematics. If we agree to denote by "$\circ$" all pre-bifurcation fields, we write $w\to\mathring{w}+w$ and $\boldsymbol{N}\to\mathring{\boldsymbol{N}}+\boldsymbol{N}$ in (1), where $(\mathring{w},\mathring{\boldsymbol{N}})$ represents the basic state and $(w,\boldsymbol{N})$ are the infinitesimal increments of the transverse displacement and membrane tensor, respectively (since the latter field depends on the in-plane displacement, it is tacitly assumed that $u_1\to\mathring{u}_1+u_1$ and $u_2\to\mathring{u}_2+u_2$, although $\mathring{u}_1$ and $\mathring{u}_2$ will not enter directly in our final bifurcation equations). On substituting these into (1) and subtracting the corresponding equation for $(\mathring{w},\mathring{\boldsymbol{N}})$, after ignoring the nonlinear terms in w and $\boldsymbol{N}$, it is found that

$$D\nabla^4 w-\nabla\cdot(\mathring{\boldsymbol{N}}\cdot\nabla w)-\nabla\cdot(\boldsymbol{N}\cdot\nabla\mathring{w})=0, \tag{8a}$$

$$\nabla\cdot\boldsymbol{N}=\boldsymbol{0}. \tag{8b}$$

Since both $\mathring{\boldsymbol{N}}$ and $\boldsymbol{N}$ are divergence-free, Eq. (8a) can be cast in the more convenient form

$$D\nabla^4 w-\mathring{\boldsymbol{N}}:(\nabla\otimes\nabla w)-\boldsymbol{N}:(\nabla\otimes\nabla\mathring{w})=0. \tag{9}$$

Before we attempt to simplify the left-hand side further, $\boldsymbol{N}$ has to be expressed in terms of the components of the pre-buckling and incremental displacement fields. After routine calculations the increment of $\boldsymbol{N}$ given in (7) becomes

$$\boldsymbol{N}=C\left\{\left(\frac{1-\nu}{2}\right)[\nabla\otimes\boldsymbol{u}+\boldsymbol{u}\otimes\nabla+(\nabla\mathring{w})\otimes(\nabla w)+(\nabla w)\otimes(\nabla\mathring{w})]+\nu[\nabla\cdot\boldsymbol{u}+(\nabla\mathring{w})\cdot(\nabla w)]\boldsymbol{I}\right\}. \tag{10}$$

Next, replacing (10) into (9) leads to a fourth-order partial differential equation involving only the displacement field $\boldsymbol{u}+w\boldsymbol{g}^3$. Some more simplifications are still possible – for the sake of brevity, we state only the final result here,

$$D\nabla^4 w-(\nabla\otimes\nabla w):\boldsymbol{A}^{(1)}-(\nabla\otimes\boldsymbol{u}):\boldsymbol{A}^{(2)}-(\nabla w)\cdot\boldsymbol{a}^{(1)}-(\nabla\cdot\boldsymbol{u})\,b^{(1)}=0, \tag{11}$$

where the coefficients

$$\boldsymbol{A}^{(1)}:=\mathring{\boldsymbol{N}},\quad \boldsymbol{A}^{(2)}:=C(1-\nu)\nabla\otimes\nabla\mathring{w},\quad b^{(1)}:=C\nu(\nabla^2\mathring{w}),$$

$$\boldsymbol{a}^{(1)}:=C[(1-\nu)(\nabla\otimes\nabla\mathring{w})\cdot\nabla\mathring{w}+\nu(\nabla^2\mathring{w})\nabla\mathring{w}]$$

will be available after solving the pre-buckling problem (3) and (4).

Transforming (8b) in terms of displacements is somewhat more involved, but in principle a fairly straightforward task. Skipping the steps involved in these calculations, the final result reads

$$\nabla^2\boldsymbol{u}+\left(\frac{1+\nu}{1-\nu}\right)\nabla(\nabla\cdot\boldsymbol{u})+(\nabla\otimes\nabla w)\cdot\boldsymbol{a}^{(2)}+(\nabla w)\cdot\boldsymbol{A}^{(3)}+(\nabla^2 w)\boldsymbol{a}^{(3)}+(\nabla w)\,b^{(2)}=\boldsymbol{0}, \tag{12}$$

where

$$\boldsymbol{A}^{(3)}:=\left(\frac{1+\nu}{1-\nu}\right)\nabla\otimes\nabla\mathring{w},\quad \boldsymbol{a}^{(2)}:=\left(\frac{1+\nu}{1-\nu}\right)\nabla\mathring{w},\quad \boldsymbol{a}^{(3)}:=\nabla\mathring{w},$$

$$b^{(2)}:=\nabla^2\mathring{w}.$$

There are different ways in which these equations can be non-dimensionalised. Motivated by the pre-buckling solution described in Section 2, we use the rescaled variables/parameters recorded in (2) together with $\overline{u}_j := (u_j/h)(a/h)$ for $j = 1, 2$. To simplify the calculations further, let us introduce $\boldsymbol{\theta}$ according to $\nabla \overset{\circ}{w} = (d\overset{\circ}{w}/d\rho)\boldsymbol{e}_r = \Theta \boldsymbol{e}_r =: \boldsymbol{\theta}$ and notice that $\nabla \otimes \nabla \overset{\circ}{w} = \nabla \otimes \boldsymbol{\theta} \equiv (d\Theta/d\rho)\boldsymbol{e}_r \otimes \boldsymbol{e}_r + (\Theta/\rho)\boldsymbol{e}_\theta \otimes \boldsymbol{e}_\theta$. On substituting the non-dimensional variables into the partial differential equations (11) and (12), they become

$$\frac{1}{12}\nabla^4 w - (\mu^2 \boldsymbol{I} + \widehat{\boldsymbol{N}}) : (\nabla \otimes \nabla w) - (\nabla \otimes \boldsymbol{u}) : [(1-\nu)\nabla \otimes \boldsymbol{\theta}]$$
$$-(\nabla w)\cdot[(1-\nu)(\nabla \otimes \boldsymbol{\theta})\cdot\boldsymbol{\theta} + \nu(\nabla\cdot\boldsymbol{\theta})\boldsymbol{\theta}] - (\nabla\cdot\boldsymbol{u})[\nu(\nabla\cdot\boldsymbol{\theta})] = 0, \quad (13)$$

$$\nabla^2 \boldsymbol{u} + \left(\frac{1+\nu}{1-\nu}\right)\nabla(\nabla\cdot\boldsymbol{u}) + (\nabla\otimes\nabla w)\cdot\left[\left(\frac{1+\nu}{1-\nu}\right)\boldsymbol{\theta}\right]$$
$$+(\nabla w)\cdot\left[\left(\frac{1+\nu}{1-\nu}\right)\nabla\otimes\boldsymbol{\theta}\right] + (\nabla^2 w)\boldsymbol{\theta} + (\nabla w)(\nabla\cdot\boldsymbol{\theta}) = \boldsymbol{0}, \quad (14)$$

where we have dropped the horizontal bar for notational simplicity.

The solution of these equations is sought by using functions with separable variables, i.e.

$$u_r(\rho,\theta) = U(\rho)\cos m\theta, \qquad u_\theta(\rho,\theta) = V(\rho)\sin m\theta,$$
$$w(\rho,\theta) = W(\rho)\cos m\theta. \quad (15)$$

with the arbitrary integer $m \geq 0$ being determined subject to the requirement that it renders the global minimum of the curve $\lambda = \lambda(m)$. The amplitudes U, V, and W satisfy the following non-dimensional coupled system of three ordinary differential equations:

$$\mathcal{A}_{112}U'' + \mathcal{A}_{111}U' + \mathcal{A}_{110}U + \mathcal{A}_{121}V' + \mathcal{A}_{120}V + \mathcal{A}_{132}W''$$
$$+\mathcal{A}_{131}W' + \mathcal{A}_{130}W = 0, \quad (16a)$$

$$\mathcal{A}_{211}U' + \mathcal{A}_{210}U + \mathcal{A}_{222}V'' + \mathcal{A}_{221}V' + \mathcal{A}_{220}V + \mathcal{A}_{231}W'$$
$$+\mathcal{A}_{230}W = 0, \quad (16b)$$

$$\mathcal{A}_{311}U' + \mathcal{A}_{310}U + \mathcal{A}_{320}V + \mathcal{A}_{334}W'''' + \mathcal{A}_{333}W''' + \mathcal{A}_{332}W''$$
$$+\mathcal{A}_{331}W' + \mathcal{A}_{330}W = 0, \quad (16c)$$

where the dash indicates differentiation with respect to ρ and the labelling of the coefficients $\mathcal{A}_{ijk}$ is done by observing the following convention. The first index i refers to the number of the equation in which it appears, the second corresponds to the place of the unknown function in the sequence (U, V, W), and k indicates the order of differentiation on the function associated with that coefficient. For the sake of completeness, the definitions of these coefficients are recorded below,

$$\mathcal{A}_{112} = 1,$$
$$\mathcal{A}_{111} = \frac{1}{\rho},$$
$$\mathcal{A}_{110} = -\frac{1}{\rho^2}\left[1 + \frac{1}{2}(1-\nu)m^2\right],$$
$$\mathcal{A}_{121} = \frac{1}{2\rho}(1+\nu)m,$$
$$\mathcal{A}_{211} = -\frac{1}{2\rho}(1+\nu)m,$$
$$\mathcal{A}_{210} = -\frac{1}{2\rho^2}(3-\nu)m,$$
$$\mathcal{A}_{222} = \frac{1}{2}(1-\nu),$$
$$\mathcal{A}_{221} = \frac{1}{2\rho}(1-\nu),$$
$$\mathcal{A}_{311} = -\left[\frac{d\Theta}{d\rho} + \nu\left(\frac{\Theta}{\rho}\right)\right],$$
$$\mathcal{A}_{310} = -\frac{1}{\rho}\left[\nu\left(\frac{d\Theta}{d\rho}\right) + \frac{\Theta}{\rho}\right],$$
$$\mathcal{A}_{332} = -\frac{1}{12\rho^2}(2m^2+1) - \left(\mu^2 + \widehat{N}_{rr}\right),$$
$$\mathcal{A}_{331} = \frac{1}{12\rho^3}(2m^2+1) - \frac{1}{\rho}\left(\mu^2 + \widehat{N}_{\theta\theta}\right) - \left[\left(\frac{d\Theta}{d\rho}\right)\Theta + \nu\left(\frac{\Theta^2}{\rho}\right)\right],$$
$$\mathcal{A}_{330} = \frac{1}{12\rho^4}m^2(m^2-4) + \frac{m^2}{\rho^2}\left(\mu^2 + \widehat{N}_{\theta\theta}\right).$$

$$\mathcal{A}_{120} = -\frac{1}{2\rho^2}(3-\nu)m,$$
$$\mathcal{A}_{132} = \Theta,$$
$$\mathcal{A}_{131} = \left[\frac{d\Theta}{d\rho} + \frac{\Theta}{\rho}(1-\nu)\right],$$
$$\mathcal{A}_{130} = -\frac{\Theta}{2\rho^2}(1-\nu)m^2$$
$$\mathcal{A}_{220} = -\frac{1}{\rho^2}\left[m^2 + \frac{1}{2}(1-\nu)\right],$$
$$\mathcal{A}_{231} = -\frac{\Theta}{2\rho}(1+\nu)m,$$
$$\mathcal{A}_{230} = -\frac{1}{2\rho}\left(\frac{d\Theta}{d\rho} + \frac{\Theta}{\rho}\right)(1-\nu)m,$$
$$\mathcal{A}_{333} = \frac{1}{6\rho},$$
$$\mathcal{A}_{320} = m\mathcal{A}_{310}, \qquad \mathcal{A}_{334} = \frac{1}{12},$$

A number of boundary constraints can be considered. We shall assume that in the transverse direction clamped boundary conditions apply, i.e. $w(r,\theta) = (\partial w/\partial r)(r,\theta) = 0$ at $r = a$. In addition, the in-plane constraints amount to prescribing the radial stress distribution along the circumference of the membrane. Thus, it transpires that $N_{rr}(r,\theta) = N_{r\theta}(r,\theta) = 0$ at $r = a$. Using (10) and the separable-variable solution (15), it transpires that

$$W(1) = W'(1) = 0, \tag{17a}$$

$$U'(1) + \nu U(1) + \nu m V(1) = 0, \quad mU(1) + V(1) - V'(1) = 0. \tag{17b}$$

We also have the regularity conditions at the centre of the plate,

$$U(0) = V(0) = W(0) = W'(0) = 0. \tag{18}$$

4. Numerical results

The numerical simulations of the problem consisting of (16) together with (17) and (18) was carried out by two different methods in order to ensure the correctness of the results obtained. In either case the pre-buckling equations (3) must be solved at the same time. The first approach was based on a classical imperfection/perturbation strategy (e.g., see Timoshenko and Gere, 1961; Jillella and Peddieson, 2012) which required casting our problem as

$$\frac{d\boldsymbol{w}}{d\rho} = \boldsymbol{A}(\rho; P, \mu)\boldsymbol{w} + \boldsymbol{f}, \qquad 0 < \rho < 1, \tag{19}$$

by introducing the dependent variables $w_1 := U$, $w_2 := U'$, $w_3 := V$, $w_4 := V'$, $w_5 := W$, $w_6 := W'$, $w_7 := W''$, $w_8 := W'''$, $w_9 := N_{rr}$, $w_{10} := N_{\theta\theta}$, $w_{11} := \Theta$, $w_{12} := \Theta'$ and then letting $\boldsymbol{w} := [w_1, w_2, \ldots, w_{12}]^T$. The expression of the matrix $\boldsymbol{A} \in M_{12\times 12}(\mathbb{R})$ can be easily found by carrying out the above substitutions; the boundary conditions admit a similar representation in terms of w_j ($j = 1, 2, \ldots, 12$). Here $\boldsymbol{f} \in M_{12\times 1}(\mathbb{R})$ is a suitably chosen (arbitrary) constant vector. By fixing all parameters in the problem except P, the eigenvalue is determined by gradually increasing this quantity from 0 until a critical value is reached, P^* (say), such that the L^2-norm of $\boldsymbol{w}$ becomes very large as $P \to P^*$, $P < P^*$; this is the smallest positive eigenvalue. Once this value has been found one can use numerical continuation techniques to identify the critical mode number or study the influence of other parameters on P^*. A more orthodox route is to set $\boldsymbol{f} = \boldsymbol{0}$ in (19) right from the outset and then solve the resulting eigenvalue problem using traditional methods (e.g., finite differences or spectral methods). More details of the numerical integration strategies that led to the result obtained in this section will appear in the forthcoming work by Coman (in preparation). Here we mention in passing that the second approach just mentioned is not entirely a routine affair because of the singular character of the original eigenproblem.

In Fig. 2 we use the dependence of P on $m \in \mathbb{R}$ to identify the critical load values, P_C, for four different levels of initial stretching; these correspond to the minima indicated on each one of the curves shown there. It can be seen that the critical load for $\mu = 0$ (no initial tension) is the smallest. By increasing μ the critical P gets larger, unlike the situations described in the related studies (Coman, 2008; Coman and Bassom, 2007a,b). Intuitively, this is to be expected because a highly stretched plate will oppose more resistance to the applied pressure. The number of wrinkles, m_C, follows a similar trend, suggesting the existence of a short-wavelength deformation pattern for large-μ values.

The critical eigenmodes corresponding to the points marked as S_μ are illustrated in Fig. 3. Even for the case of no initial tension, the eigenmodes are still localised, as found by Adams (1993) for the case of a plate subjected to a concentrated load at the centre. However, unlike in that work, the components V and W have no strong oscillatory component. The concentration of the eigenmodes

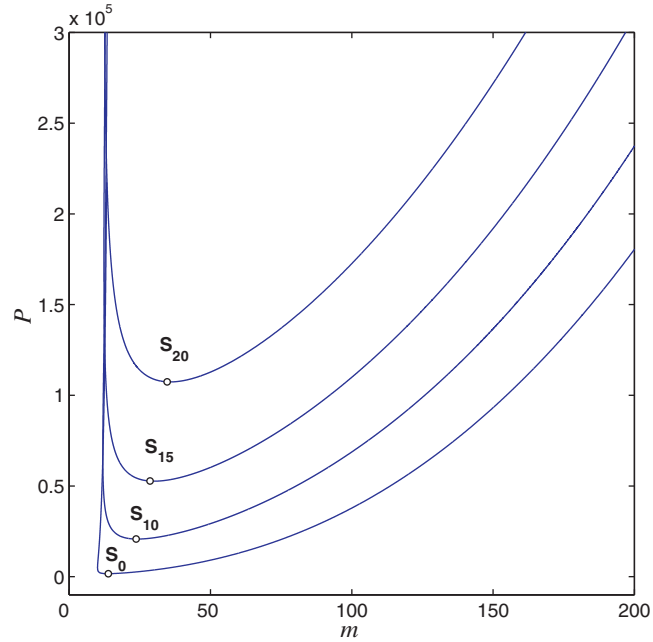

Fig. 2. Dependence of the eigenvalues P on the mode number m for different values of the background tension, μ. The points S_μ (μ=0, 10, 15, and 20) are the minima of the corresponding curves, identifying the physically relevant values of the parameters P and m. Here $\nu = 0.33$.

near the outer rim is controlled by P rather than μ although, as shown by Coman (in preparation), an asymptotic description of this phenomenon does require making precise assumptions on the size of this parameter in terms of P, which somewhat limits the practical applicability of the analysis.

We also include in Fig. 4 the behaviour of the corresponding radial and azimuthal prebuckling stresses together with the deflection slope as a function of the radial coordinate. The maximum amplitudes of $\overset{\circ}{N}_{rr}$ and $\overset{\circ}{N}_{\theta\theta}$ are monotonic increasing functions of μ; the minima of the curves in the rightmost window have the opposite property. As already found in earlier studies (Bromberg, 1956; Sheplak and Dugundji, 1998; Tong and Huang, 2002) bending effects tend to be confined to a small boundary layer near $\rho = 1$ as $\mu \gg 1$. It is worth pointing out that, as evidenced by the curves shown in the middle window of Fig. 4, the region of compressive stresses adjacent to the outer edge of the plate tends to shrink (albeit very slowly) as the initial tension increases. So in this respect the situation is analogous to the stretched annular plates analysed by the author and his associates (e.g., see Coman, 2008; Coman and Bassom, 2007a,b).

It is known that a pressurised thin plate exhibits two distinct types of behaviour: for small pressures bending effects are dominant and this corresponds to the almost straight parts of the curves shown in Fig. 5. These effects become negligible as the pressure gets stronger and subsequently the system behaves like a membrane (see Sheplak and Dugundji, 1998). Superimposed on the curves seen in that Figure are a number of white markers which coincide with the onset of wrinkling. It is clear that by the time this occurs the prebuckling state of stress is close to the membrane regime. It can be shown that the bending effects are confined to a thin boundary layer of size $\mathcal{O}(P^{-1/3})$ and that the localisation process witnessed in Fig. 3 takes place within an $\mathcal{O}(P^{-1/6})$ layer ($P \gg 1$, $\mu \neq 0$). Routine scaling arguments presented in Coman (in preparation) also indicate that $m_C = \mathcal{O}(P^{1/4})$, and a strategy for finding the corresponding prefactor is proposed therein.

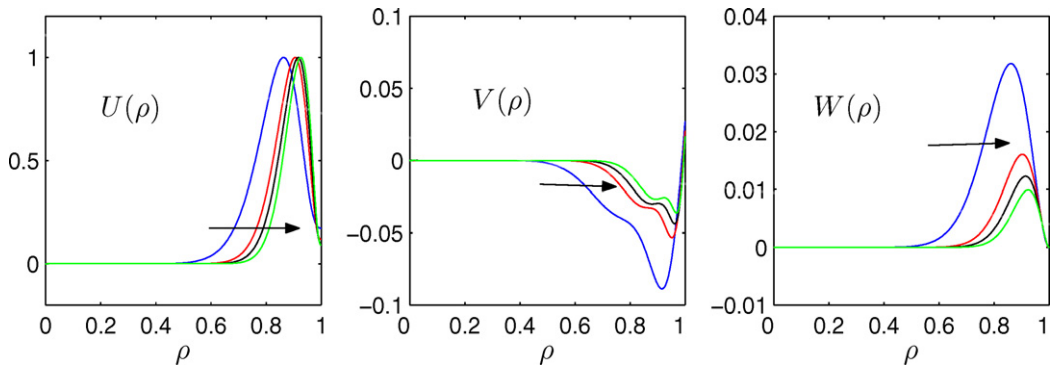

Fig. 3. The critical eigenmodes associated with the points S_μ in Fig. 2; in each window the leftmost curve represents the solution corresponding to $\mu = 0$ and the arrow shows the direction in which μ increases. The eigenmodes are normalised by the maximum amplitude of U.

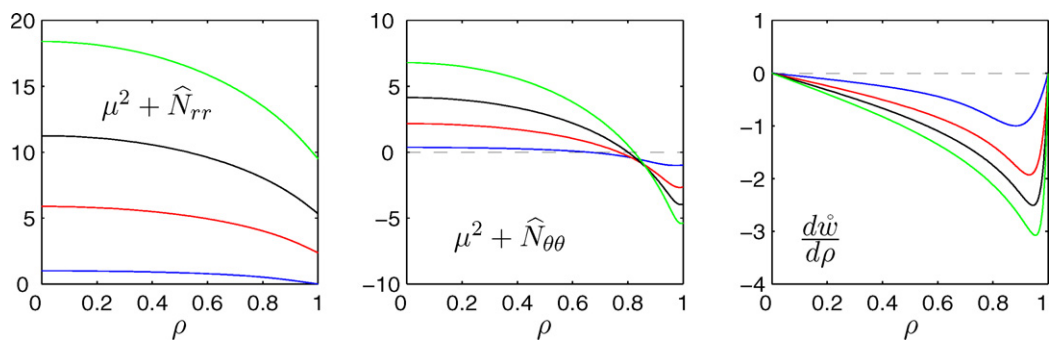

Fig. 4. Normalised prebuckling fields for the eigenmodes shown in Fig. 3. In all windows the reference curves are those corresponding to $\mu = 0$; the maximum amplitudes for $\mathring{N}_{rr}$ and $\mathring{N}_{\theta\theta}$ increase monotonically with μ, while in the rightmost window the minima of the curves shown there have exactly the opposite property.

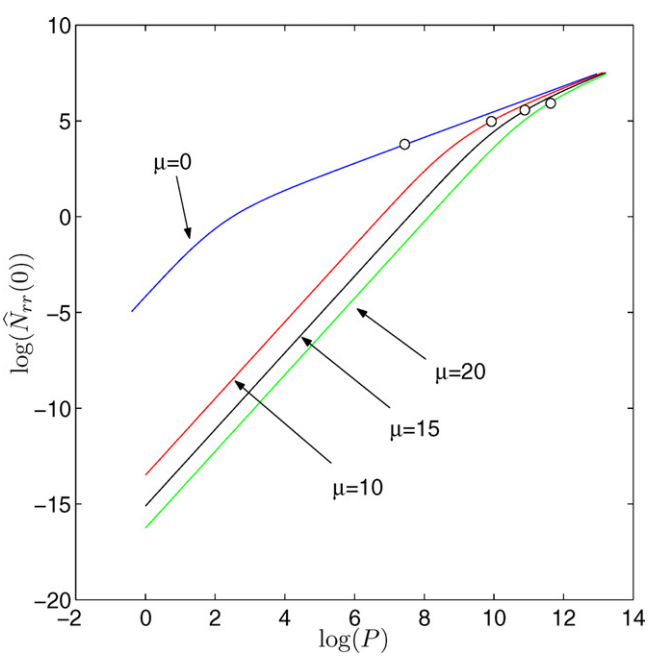

Fig. 5. The dependence of the centre prebuckling radial stress on the loading for several values of in-plane tension. The markers correspond to the onset of wrinkling as predicted by solving the boundary-value problem (16)–(18).

5. Conclusions

We have investigated the edge-buckling experienced by a stretched circular thin plate subjected to uniform transverse pressure. In particular, a special attention has been payed to the role of the background tension on the localised character of the lowest eigenmode. A coordinate-free system of bifurcation equations has been derived and later specialised to a circular geometry (although similar boundary-value problems can be immediately obtained for rectangular or elliptical plates). We showed that the critical load tends to increase with the background tension and the eigenmodes become progressively more focused towards the outer rim of the plate. Also, the mode number was found to increase in a similar fashion. This numerical evidence suggests that unlike the simpler situations explored by Coman (2008) and Coman and Bassom (2007a,b), here the localisation is controlled by the size of the eigenvalue. The details of the strategy required for capturing the leading order asymptotics of this scenario will be presented in a forthcoming study (Coman, in preparation).

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